ITA0329-MOBILE COMPUTING **K.BALAJI**

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Assessment Tool 2 – Short Answers

ANSWERS:

Q1. Full-duplex voice channels

The total spectrum is 45 MHz, and each simplex channel requires 25 kHz. A full-duplex channel requires two simplex channels.

Number of full-duplex channels = 45,000 ÷ (25 × 2) = 900.

Answer: 900 full-duplex voice channels.

Q2. Channels per cell

The network has 560 channels shared among 7 cells.

Channels per cell = 560 ÷ 7 = 80.

Answer: Each cell gets 80 channels.

Q3. GSM simultaneous voice channels

Each carrier supports 8 time slots, and there are 100 carriers.

Total voice channels = 100 × 8 = 800.

Answer: 800 simultaneous voice channels.

Q4. Power ratio in decibels

The base station transmits 80 W, while the handset transmits 4 W.

Power ratio = $10 \log10(80 \div 4) = 10 \log_{10}(20)$.

Answer: Approximately 13 dB.

Q5. Carrier-to-interference ratio

The serving signal is −60 dBm, and the interference is −92 dBm.

C/I = −60 − (−92) = 32 dB.

Answer: 32 dB.

Q6. Channels allocated to each cell

Total channels = 42,000 kHz ÷ 300 kHz = 140.

Channels per cell = 140 ÷ 7 = 20.

Answer: 20 channels per cell.

Q7. Received power after doubling the distance

The initial received power is 64 W, and the path-loss exponent is 4.

$$\text{New power} = 64 \div 2^4 = 64 \div 16 = 4 \text{ W}.$$

Answer: 4 W.

Q8. Transmit power when the cell radius doubles

According to the inverse-square law, the required power increases with the square of the distance.

$$\text{Power factor} = 2^2 = 4.$$

Answer: The transmit power should be increased to 4 times its original value.

Q9. Total channels in a cluster

The cluster contains 4 cells, with 15 channels in each cell.

$$\text{Total channels} = 4 \times 15 = 60.$$

Answer: 60 channels.

Q10. Doppler shift

The speed is 100 km/h, and the carrier frequency is 900 MHz.

$$\text{Speed in m/s} = 100 \div 3.6 = 27.78 \text{ m/s}.$$

$$\text{Doppler shift} = (27.78 \times 900 \times 10^6) \div (3 \times 10^8) \approx 83.3 \text{ Hz}.$$

Answer: Approximately 83.3 Hz for motion directly toward or away from the transmitter.

Q11. Duration of one TDMA time slot

The frame duration is 5 ms and contains 8 equal time slots.

$$\text{Slot duration} = 5 \div 8 = 0.625 \text{ ms}.$$

Answer: 0.625 milliseconds.

Q12. Number of carrier frequencies

The allocated bandwidth is 24 MHz, and each carrier uses 200 kHz.

$$\text{Number of carriers} = 24{,}000 \div 200 = 120.$$

Answer: 120 carrier frequencies.

Q13. Received power in dBm

The handset transmits at 8 W, and the path loss is 33 dB.

Transmit power = 10 log10(8,000) =39.03 dBm.

Received power = 39.03 − 33 = 6.03 dBm.

Answer: Approximately 6.03 dBm.

Q14. Free-space path loss

The frequency is 1,900 MHz, and the distance is 2.5 km.

FSPL = 32.44 + 20 log10(1900) + 20 $\log_{10}$(2.5).

Answer: Approximately 106 dB.

Q15. Total call capacity of a cluster

There are 6 cells, and each cell can handle 250 simultaneous calls.

Total capacity = 6 × 250 = 1,500.

Answer: 1,500 simultaneous calls.

Q16. Number of active calls

The system processes 4,000 calls per second, and the average call lasts 3 minutes.

Call duration = 3 × 60 = 180 seconds.

Active calls = 4,000 × 180 = 720,000.

Answer: Approximately 720,000 active calls, assuming the call rate remains constant.

Q17. Total bandwidth used

There are 150 channels, each occupying 25 kHz.

Total bandwidth = 150 × 25 = 3,750 kHz.

Answer: 3.75 MHz.

Q18. Total guard-band bandwidth

Assuming there are 30 guard bands, each measuring 40 kHz:

Total bandwidth = 30 × 40 = 1,200 kHz.

Answer: 1.2 MHz.

Q19. Approximate number of handoffs

The cell radius is 1.5 km, so the diameter is 3 km. The vehicle travels 90 km in one hour.

Number of handoffs = 90 ÷ 3 = 30.

Answer: Approximately 30 handoffs per hour.

Q20. Total user-handling capacity

There are 8 regional centers, each managing 40 base stations. Each base station serves 250 users.

Total capacity = $8 \times 40 \times 250 = 80{,}000$.

Answer: 80,000 users.

Q21. Number of cells required

The service region covers 400 km², and the cell radius is 400 m or 0.4 km.

Using the given approximation:

Cell area = $R^2 = 0.4^2 = 0.16$ km².

Number of cells = $400 \div 0.16 = 2{,}500$.

Answer: Approximately 2,500 cells.

Q22. Number of FDMA channels

The total spectrum is 6 MHz, and each channel requires 30 kHz.

Number of channels = $6{,}000 \div 30 = 200$.

Answer: 200 channels.

Q23. Offered traffic in Erlangs

There are 1,200 subscribers. Each makes 4 calls per hour, with an average duration of 2.5 minutes.

Traffic = $(1{,}200 \times 4 \times 2.5) \div 60 = 200$ Erlangs.

Answer: 200 Erlangs.

Q24. Maximum number of simultaneous users

The allocated bandwidth is 40 MHz, and each user requires 400 kHz.

Number of users = $40{,}000 \div 400 = 100$.

Answer: 100 simultaneous users.

Q25. Total network user capacity

The network contains 1,200 base stations, and each station supports 600 simultaneous users.

Total capacity = $1{,}200 \times 600 = 720{,}000$.

Answer: 720,000 simultaneous users.